EXPLAINER · CLUSTER C · CONTEXTUAL STOCHASTIC OPTIMIZATION

한국어 ← Research Map

Contextual SO in Pictures — C2 · C3 · C4

In one sentence: **“If you have observed the city features X, the decision should depend on X too.”** C2 gives the map of methods, C3 the recipe for turning X into scenarios, and C4 the insurance for when those scenarios cannot be trusted.

`X = city features (income · density · POI · zero-car households …)` `D = robotaxi demand` `q = vehicles allocated`

`c(q,D) = c⁺(q−D)⁺ + c⁻(D−q)⁺`

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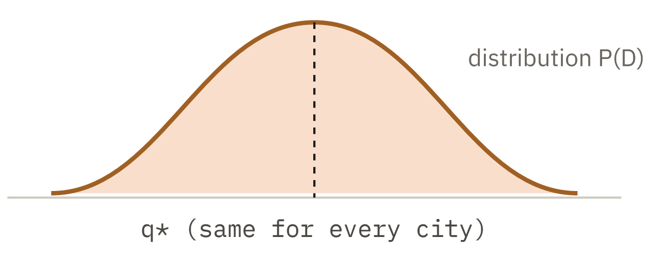
00 Core idea — from “one average” to “a different distribution per city”

BEFORE · Unconditional (Phase 1)

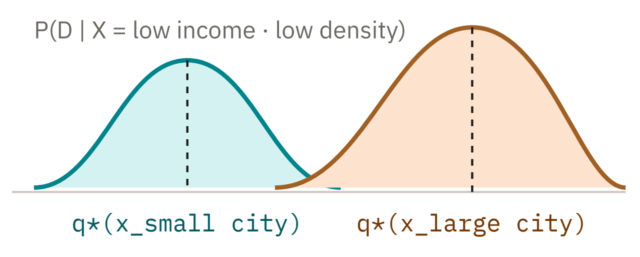
$\min_q E[\, c(q, D)\,]$



All cities share one demand distribution and one fleet size. Atlanta and Chicago get the same answer.

AFTER · Conditional = Contextual (Phase 2)

$\min_q E[\, c(q, D) \mid X = x\,] \rightarrow q^*(x)$



Observe the city's features x first, then size the fleet for that city's **conditional distribution**. The decision becomes a function q*(x) — a **policy**.

The hard part

P(D | X = x) is never observed directly. All we have are pairs from past cities $(x_1,D_1), \dots, (x_n,D_n)$. **“How do we build the conditional distribution for a new city x from these pairs?”** is the single question behind C2 · C3 · C4.

C2 Sadana et al. 2024 Survey — the map of methods

Methods classified by “how tightly prediction and optimization are coupled”

◀ Separate (SLO · Sequential) Coupled (ILO · Integrated) ▶

① Estimate-then-Optimize

```
X → regression / density est.
→ P̂(D|x)
→ min E_P̂[c]
```

Build the prediction model first, then **plug** its output into the optimization. Most intuitive and modular; standard statistical tools apply directly.

C3 ER-SAA lives here

② Weighted SAA

```
w_j(x) = similarity(x, x_j)
min Σ_j w_j(x) c(q, D_j)
```

No distribution is estimated; **past cities that resemble the new one** get larger weights. kNN · kernel · random-forest weights. Same idea as nonparametric regression.

Phase 2 alternative

③ Decision-Focused Learning

```
X → f_θ → optimize → q
→ update θ with decision loss
```

Train the predictor on **final cost**, not MSE. Hard because you must differentiate through the optimization, but prediction is aligned with the decision.

④ Direct Policy Learning

```
learn q = π_θ(x) directly
min Σ_j c(π_θ(x_j), D_j)
```

Skip demand prediction and learn the **decision rule itself**. Wins when the optimal decision is simpler than the distribution. Hard to interpret.

C2's key claim · prediction accuracy ≠ decision quality

	Model A	Model B
MSE (prediction error)	low ✓	high
Where it errs	**under**-predicts in large cities	over-predicts in small cities
Realized cost ($c^- \gg c^+$)	high — heavy unmet demand	low ✓

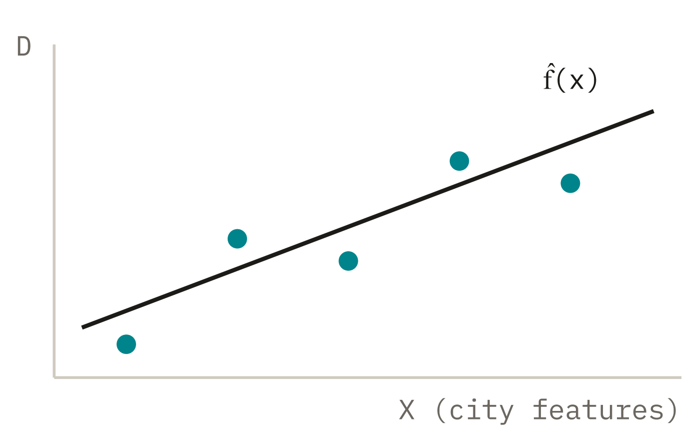
When the shortage penalty c^- dwarfs the idle penalty c^+, *where* you err matters more than *how much*. So Phase 2 should be scored on **regret · unmet demand · realized cost**, not MSE.

C3 Kannan · Bayraksan · Luedtke — ER-SAA (Empirical Residuals SAA)

“Regression carries the pattern, residuals carry the scatter”

With only n past cities you cannot estimate the conditional distribution at a new x directly. ER-SAA's fix: split demand as **D = f(X) + ε**, learn f by regression, and **reuse the residuals as-is** for the distribution of ε.

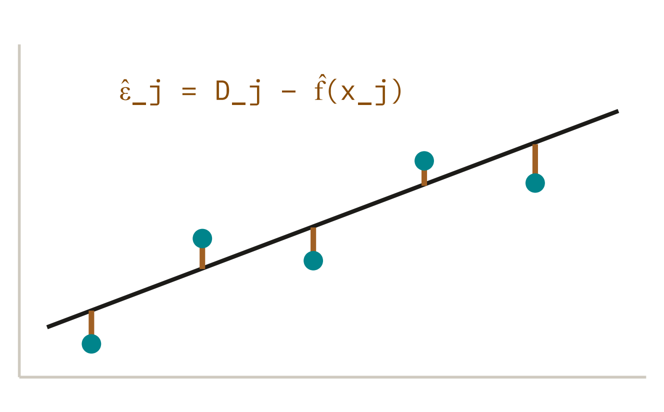
1 Fit a regression



`(x_j, D_j), j=1..n → f̂ = argmin Σ (D_j − f(x_j))²`

Any regression works — OLS, Lasso, random forest. From 5 cities, learn the trend “features → mean demand”.

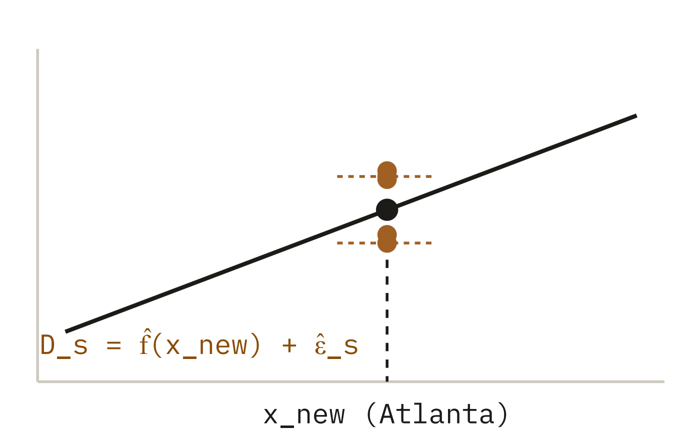
2 Collect the residuals



`{ε̂_1, …, ε̂_n} = distances from the fitted line`

Key assumption: the scatter ε has the **same distribution regardless of X**. So any city's residual is a valid sample of “possible scatter”.

3 Attach residuals to the new city



`scenarios s=1..n: D_s(x_new) = f̂(x_new) + ε̂_s`

“Copy-paste” the residuals around the regression prediction (the dot). n training cities = n scenarios.

4 Decide with SAA

```
q̂(x_new) = argmin_q
  (1/n) Σ_s c(q, f̂(x_new) + ε̂_s)
```

Exactly ordinary SAA — only the scenarios have been shifted to x_new. For a newsvendor the answer is the **$c^-/(c^+ + c^-)$ quantile** of the scenarios.

The paper's contribution: if f̂ is consistent, q̂(x) converges to the true q*(x) (consistency + rates).

Walking through the numbers (illustrative values · unit: 100 rides/day)

① 5 training cities → f̂

city	f̂(x_j)	D_j	ε̂_j
SF	14	15.5	+1.5
PHX	9	8.0	−1.0
LA	18	17.2	−0.8
AUS	8	9.1	+1.1
MIA	11	10.2	−0.8

② New city ATL: f̂(x_ATL) = 10 → 5 scenarios

```
D_s = 10 + ε̂_s
= {11.5, 9.0, 9.2, 11.1, 9.2}
sorted: 9.0 · 9.2 · 9.2 · 11.1 · 11.5
```

③ Newsvendor decision

```
c⁺=1 (idle), c⁻=3 (lost to human ride-hailing)
target quantile = 3/(1+3) = 0.75
→ q̂(x_ATL) ≈ 11.1
```

In Phase 1 (one pooled distribution) ATL would have received the same answer as SF. Now ATL gets its own.

Why it's good No need to estimate a full conditional distribution — **one regression + residuals** is enough. Works with small n (5–20 cities). Unlike kNN, regression extrapolates even when no “similar city” exists.

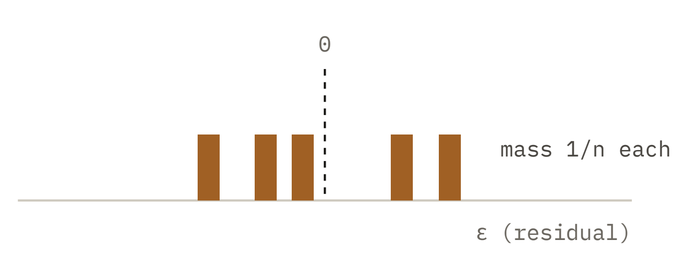
Caveats Assumes residuals are independent of X. If large cities fluctuate more (**heteroscedasticity**), extend to D = f(X) + Q(X)·ε — standardize residuals by city scale before reuse. And if f̂ itself is wrong, every scenario is biased → C4.

C4 Kannan · Bayraksan · Luedtke — Residuals-based DRO

“Don't trust the residual distribution either — hedge against the worst case nearby”

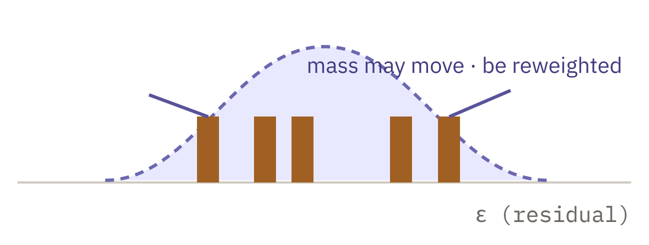
C3's scenarios can be wrong for two reasons: (i) with only n residuals, the empirical distribution differs from the true ε distribution; (ii) if f̂ is off, the whole center shifts. C4 places an **ambiguity set** around the empirical residual distribution and optimizes against the worst distribution inside it.

C3 · ER-SAA — trusts the single empirical P̂_n



Five residuals = five “possible worlds”. Any unobserved large residual (e.g. a sudden demand spike) gets probability zero.

C4 · Residuals-DRO — worst case within radius ε of P̂_n



Only assume “the true distribution is within distance ε of P̂_n”. Hedge against the **costliest** distribution obtained by nudging residuals (Wasserstein) or reweighting them (φ-divergence).

Formula — one term changes from C3

```
C3  min_q  E_{P̂_n}[ c(q, f̂(x) + ε) ]

C4  min_q  sup_{Q : d(Q, P̂_n) ≤ ε_n} E_Q[ c(q,
f̂(x) + ε) ]
```

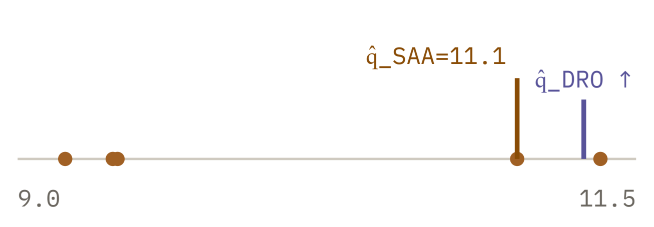
d = Wasserstein distance or φ-divergence. The radius ε_n → 0 as n grows, so C4 collapses to C3. The relationship between the two papers is visible in the formula itself.

What the radius ε_n absorbs

- **Sampling error** — the empirical distribution of n residuals ≠ true ε distribution. Smaller n → larger ε_n.
- **Regression error** — f̂ ≠ f. The paper's core contribution: how to fold f̂'s estimation error into the radius while preserving consistency.
- **Heteroscedasticity** — in D = f(X)+Q(X)ε, Q̂ can also be wrong → build the ambiguity set on standardized residuals.

Phase 3 in practice: calibrate ε_n by cross-validation / bootstrap and compare SAA vs DRO out-of-sample cost (B2 logic).

Effect on the decision (newsvendor intuition)



With $c^- > c^+$ the worst-case distribution pushes mass toward high demand → DRO **allocates more vehicles** than SAA. The radius is the “amount of conservatism”; the price is more idle vehicles on average.

Σ The three papers on one line

Question	C2 · Survey	C3 · ER-SAA	C4 · Residuals-DRO
How should X enter the decision?	**Map.** 4 approaches · SLO/ILO · evaluate by decision quality	Estimate-then-optimize (SLO)	Same as C3 + robustification
Concretely, how to go X → scenarios?	—	**Recipe.** D_s = f̂(x) + ε̂_s, then plain SAA	Same as C3
What if the scenarios are wrong?	Proposes the Contextual DRO concept	—	**Insurance.** Radius-ε_n ambiguity set around the empirical residuals
Place in our roadmap	Rationale for Phase 2 design · evaluation metrics	**Phase 2** implementation	**Phase 3** implementation

One-liners

C2: "Better prediction is not better decisions." - C3: "Regression carries the pattern, residuals carry the scatter." - C4: "Don't trust the residual distribution either — hedge against the worst case within the radius."